

Hospitality Revenue & Performance Analysis Dashboard Project Report

1. Project Overview

This project analyzes the revenue performance of **AtliQ Grands**, a luxury hotel chain operating across multiple cities in India. The objective is to help the revenue management team identify factors affecting revenue, occupancy, and customer satisfaction using Power BI. The dashboard provides interactive insights into hotel performance, booking trends, room utilization, and key revenue metrics to support data-driven decision-making.

2. Business Problem Statement

Domain: Hospitality

Function: Revenue Management

AtliQ Grands has been operating in the hospitality industry for over 20 years. Due to increasing competition and ineffective management decisions, the company has experienced a decline in market share and revenue within the luxury and business hotel segments.

Since the organization does not have an in-house analytics team, they hired a third-party data analyst to analyze historical booking data and provide actionable business insights.

Objectives

- Create all required revenue management KPIs.
 - Design an interactive Power BI dashboard based on stakeholder requirements.
 - Generate additional business insights beyond the provided dashboard mock-up.
 - Support management in improving occupancy, pricing strategy, and revenue performance.
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3. Dataset Summary

- **Time Period:** May 2022 – July 2022 (92 Days)
- **Data Model:** Star Schema
- **Tables:** 5 (3 Dimension Tables & 2 Fact Tables)

Dimension Tables

- **dim_date:** Contains calendar information such as Date, Month-Year, Week Number, and Day Type (Weekday/Weekend).
- **dim_hotels:** Contains hotel details including Property ID, Property Name, Category (Luxury/Business), and City.
- **dim_rooms:** Contains room classifications including Standard, Elite, Premium, and Presidential.

Fact Tables

- **fact_bookings:** Stores booking-level transactions, including booking dates, check-in/check-out dates, booking status, booking platform, guest ratings, revenue generated, and revenue realized.
 - **fact_aggregated_bookings:** Stores daily room availability and successful bookings for each property and room category, used to calculate occupancy and operational KPIs.
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4. Data Preparation & Cleaning (Power Query)

Before building the dashboard, the dataset was cleaned and transformed to ensure data quality and accurate analysis.

Data Cleaning Steps

- Imported all five tables into Power BI.
- Verified and corrected data types for all columns.
- Removed unnecessary whitespaces using **Trim**.
- Checked for null values and inconsistent records.
- Validated relationships between fact and dimension tables.
- Built a Star Schema data model for efficient reporting.

Business Logic Correction

In the hospitality industry:

- **Weekend:** Friday and Saturday
- **Weekday:** Sunday to Thursday

The original **Day Type** values in the `dim_date` table were verified and corrected to match the industry's standard business definition, ensuring accurate occupancy and revenue trend analysis.

Challenge Faced

The **Week Number** column in the `dim_date` table contained values such as:

```
W 19
W 20
W 21
```

Because the column was stored as **Text**, Power BI could not sort or analyze the data chronologically for Week-over-Week calculations.

Solution

- Removed the "**W**" prefix from the Week Number values.
 - Converted the column to **Whole Number**.
 - Successfully enabled chronological Week-over-Week trend analysis for Revenue and other KPIs.
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5. DAX Measures Created

A total of **26 business KPIs and measures** were created using DAX to monitor hotel performance, booking efficiency, and revenue management.

Revenue Metrics

- Revenue
- ADR (Average Daily Rate)
- RevPAR (Revenue Per Available Room)
- Revenue WoW Change %
- ADR WoW Change %
- RevPAR WoW Change %

Booking & Occupancy Metrics

- Total Bookings
- Total Capacity
- Total Successful Bookings
- Occupancy %
- Booking % by Platform
- Booking % by Room Class

Operational Metrics

- DBRN (Daily Booked Room Nights)
- DSRN (Daily Sellable Room Nights)
- DURN (Daily Utilized Room Nights)
- DSRN WoW Change %

Customer & Performance Metrics

- Average Rating
- Total Checked Out
- Total Cancelled Bookings
- Cancellation %
- Total No Show Bookings
- No Show Rate %
- Realisation %
- Realisation WoW Change %

Time-Based Metrics

- Number of Days

Week-over-Week Performance Metrics

- Revenue WoW Change %
- Occupancy WoW Change %
- ADR WoW Change %
- RevPAR WoW Change %
- Realisation WoW Change %
- DSRN WoW Change %

6. Power BI Dashboard

The dashboard was designed to provide management with an interactive overview of hotel performance.

Dashboard Components

KPI Cards

- Revenue → ₹1.69 Billion
- Occupancy → 58%
- Total Bookings → 133K
- RevPAR → ₹7.34K
- ADR → ₹12.70K
- Realisation → 70%

Filters (Slicers)

- City
- Room Class
 - Standard
 - Elite
 - Premium
 - Presidential

Visualizations

1. Revenue Trend (Week over Week)

- Line Chart
- Revenue by Week
- Comparison between:
 - Business Hotels
 - Luxury Hotels

Insight

Luxury hotels consistently generated higher revenue than business hotels throughout the analysis period.

2. Revenue by Category

Donut Chart

- Luxury → 61.62%
- Business → 38.38%

Insight

Luxury hotels contribute the majority of total revenue.

3. Revenue by City

Bar Chart

Revenue Performance Ranking:

- 1. Mumbai
- 2. Bangalore
- 3. Hyderabad
- 4. Delhi

Insight

Mumbai is the highest revenue-generating market, while Delhi has the lowest contribution.

Property Performance Table

The dashboard includes a detailed property-level comparison containing:

- Property ID
- Property Name
- City
- Revenue
- RevPAR
- ADR
- DBRN
- DSRN
- DURN
- Realisation %
- Cancellation %
- Average Rating

This enables stakeholders to compare hotel performance across different cities and properties.

7. Key Business Insights

- Luxury hotels generated **61.62%** of total revenue, making them the strongest revenue contributor.
- Mumbai emerged as the highest-performing city in terms of revenue generation.
- Elite room class contributed the highest revenue among all room categories.
- Overall occupancy stood at **58%**, indicating opportunities to improve room utilization.
- The hotel chain achieved a **70% Realisation Rate**, meaning a significant portion of bookings resulted in successful check-outs.
- Property-level comparison highlighted performance differences across hotels, enabling targeted operational improvements.

- Weekly revenue trends revealed fluctuations across Business and Luxury categories, helping management identify seasonal demand patterns.
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8. Business Recommendations

Increase Occupancy

Implement targeted promotional campaigns during low-demand weeks to improve room utilization.

Focus on High-Revenue Markets

Invest more marketing efforts in Mumbai and Bangalore while developing strategies to improve performance in Delhi.

Promote Elite Room Class

Since Elite rooms generate the highest revenue, introduce premium packages and upselling strategies for this category.

Reduce Booking Losses

Analyze cancellation and no-show trends to implement flexible pricing and better booking policies.

Improve Revenue Strategy

Monitor RevPAR and ADR regularly to optimize room pricing based on occupancy and seasonal demand.

Property Performance Monitoring

Use property-level KPIs to identify underperforming hotels and implement operational improvements.